
Safe Reinforcement Learning via Metric Singularities: Geodesic Policy Optimization on Reward Manifolds

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Abstract

Reward hacking—where RL agents exploit loopholes in reward specifications to achieve high reward without accomplishing intended objectives—poses a fundamental challenge for AI alignment. We propose **Singularity-Aware Geodesic Policy Optimization (SGPO)**, a geometric framework that prevents reward hacking by making dangerous regions of the reward landscape geodesically unreachable. We formalize reward spaces as Riemannian manifolds and introduce *metric singularities*: learned anisotropic metrics that assign infinite geodesic length to paths entering deceptive-reward regions, analogous to black holes in general relativity. SGPO trains a conformal metric tensor that expands distances near empirically detected danger zones, then optimizes policy updates along geodesics of the resulting manifold. On *Murky Drone Navigation*—a benchmark where a deceptive reward trap offers $10\times$ the true objective—SGPO reduces safety violations by 79% compared to PPO while maintaining $8\times$ higher returns than CPO. Topology mining on 160,800 samples from Anthropic’s HH-RLHF reveals that 30% of preference pairs lie in topologically severe risk regions, validating the practical need for geometric safety. Code and data are available in the supplementary material.

1. Introduction

Reinforcement learning from human feedback (RLHF) has become the dominant paradigm for aligning large language models with human intent (Christiano et al., 2017; Ouyang et al., 2022). Yet the reward models at its core are imperfect proxies: agents routinely discover policies

that achieve high reward through unintended behaviors—a phenomenon known as *reward hacking* (Moskovitz et al., 2024; Coste et al., 2024). Recent work demonstrates that reward hacking can emerge naturally and escalate from benign shortcuts to active sabotage (Anthropic, 2025).

Existing defenses constrain *how much* a policy deviates from a reference (KL penalties, reward clipping) or constrain *what states* are visited (constrained MDPs, control barrier functions). These approaches treat reward hacking as a scalar optimization problem. We argue that the geometric structure of reward landscapes—their topology, curvature, and connectivity—contains information that scalar methods discard.

Our approach. We formalize the reward landscape as a Riemannian manifold and introduce *metric singularities*: regions where a learned metric tensor assigns infinite geodesic distance. A policy optimizer constrained to follow geodesics of this manifold cannot reach deceptive-reward regions, regardless of the apparent reward gradient. This is analogous to black holes in general relativity: the singularity in the Schwarzschild metric prevents light from escaping, not by a force, but by the geometry of spacetime itself.

Contributions.

1. **Reward manifold formalism:** We define reward spaces as Riemannian manifolds with learned anisotropic metrics (Section 3).
2. **Metric singularity theory:** We prove that policies following geodesics on manifolds with appropriate singularities cannot enter designated danger zones, providing formal safety guarantees (Section 4, Theorem 1).
3. **SGPO algorithm:** We present Singularity-Aware Geodesic Policy Optimization, which learns the metric from data and optimizes along geodesics (Section 5).
4. **Empirical validation:** SGPO reduces violations by 79% vs. PPO and achieves $8\times$ higher returns than

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CPO on Murky Drone Navigation, while topology mining on 160,800 HH-RLHF samples reveals widespread risk structure (Section 6).

2. Background

2.1. RLHF and Reward Hacking

In RLHF, a reward model $r_\theta : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is trained from human preference comparisons, then used to optimize a policy via RL (Ouyang et al., 2022). The policy update typically maximizes $\mathbb{E}_\pi[r_\theta(s, a)] - \beta D_{\text{KL}}(\pi \| \pi_{\text{ref}})$. Reward hacking occurs when π exploits systematic errors in r_θ , achieving high proxy reward while diverging from human intent (Moskovitz et al., 2024).

2.2. Constrained MDPs and Safe RL

Constrained MDPs (CMDPs) (Altman, 1999) augment the standard objective with cost constraints: maximize $J(\pi)$ subject to $J_{C_i}(\pi) \leq d_i$. Constrained Policy Optimization (CPO) (Achiam et al., 2017) solves this via trust-region methods. While effective for known hazards, CMDPs require explicit cost functions—precisely the specification challenge that enables reward hacking.

2.3. Riemannian Geometry Preliminaries

A *Riemannian manifold* $(\mathcal{M}, g)$ is a smooth manifold equipped with a positive-definite metric tensor g that defines inner products on tangent spaces (do Carmo, 1992). The *geodesic distance* between points $p, q \in \mathcal{M}$ is:

$$d_g(p, q) = \inf_{\gamma} \int_0^1 \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt \quad (1)$$

where the infimum is over smooth curves $\gamma : [0, 1] \rightarrow \mathcal{M}$ with $\gamma(0) = p$, $\gamma(1) = q$. A key property: if g becomes singular (diverges) at a point, geodesic distance to that point becomes infinite, making it unreachable by length-minimizing paths.

3. Reward Manifold Formalism

Definition 1 (Reward Manifold). *Given a state-action embedding $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ and a reward model r_θ , the reward manifold is the Riemannian manifold $(\mathcal{M}, g_\alpha)$ where $\mathcal{M} = \phi(\mathcal{S} \times \mathcal{A}) \subseteq \mathbb{R}^d$ and g_α is a learned metric tensor parameterized by α .*

We define an *anisotropic conformal metric* that modulates the base Euclidean metric:

$$g_\alpha(x) = \Lambda_\alpha(x)^\top \Lambda_\alpha(x) \quad (2)$$

where $\Lambda_\alpha : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ is a neural network producing a (not necessarily symmetric) matrix. This general-

izes isotropic conformal metrics $g = e^{2f}I$ by allowing direction-dependent stretching.

Metric learning objective. Given a set of danger embeddings $\mathcal{D} = \{\phi(s, a) : (s, a) \in \text{danger zone}\}$ identified by anomaly detection (Section 3.1), we train Λ_α to inflate distances near $\mathcal{D}$:

$$\mathcal{L}_{\text{metric}} = \sum_{x \in \mathcal{D}} \max(0, \tau - \|\Lambda_\alpha(x)\|_F) + \lambda_{\text{reg}} \sum_{x \notin \mathcal{D}} \|\Lambda_\alpha(x) - I\|_F^2 \quad (3)$$

The first term encourages large Frobenius norm (expanded distances) near danger zones; the second regularizes toward the identity metric elsewhere, preserving the original geometry in safe regions.

3.1. Danger Zone Detection

We identify danger zones in embedding space via density-based clustering on reward model disagreement signals. Given an ensemble of reward models $\{r_{\theta_k}\}_{k=1}^K$ or a single model’s uncertainty estimates, we cluster points where:

$$\text{Var}_k[r_{\theta_k}(x)] > \epsilon \quad \text{and} \quad \max_k r_{\theta_k}(x) > \bar{r} + \sigma_r \quad (4)$$

i.e., regions of high reward *and* high disagreement—the signature of exploitable reward model errors. We use HDB-SCAN with a minimum cluster size of 3 and distance threshold $\epsilon = 0.5$ in reduced embedding space.

4. Metric Singularity Theory

The central theoretical contribution is that appropriately constructed metric singularities provide formal safety guarantees.

Definition 2 (Metric Singularity). *A metric singularity at region $\mathcal{D} \subset \mathcal{M}$ is a metric tensor g such that for any path γ entering $\mathcal{D}$:*

$$\int_{\gamma} \sqrt{g(\dot{\gamma}, \dot{\gamma})} dt = +\infty \quad (5)$$

Theorem 1 (Geodesic Safety Guarantee). *Let $(\mathcal{M}, g_\alpha)$ be a reward manifold with metric singularity at $\mathcal{D}$. If the policy update follows a geodesic of g_α from the current policy embedding $x_0 \notin \mathcal{D}$ with step size $\eta < \infty$, then $x_0 + \eta \dot{\gamma}(0) \notin \mathcal{D}$.*

Proof. By definition of geodesic distance, $d_{g_\alpha}(x_0, \mathcal{D}) = \infty$ since any path from x_0 to $\mathcal{D}$ has infinite length under g_α . A geodesic γ starting at x_0 with finite step size η traverses arc length η . Since $\eta < \infty < d_{g_\alpha}(x_0, \mathcal{D})$, the geodesic cannot reach $\mathcal{D}$. $\square$

Algorithm 1 SGPO: Singularity-Aware Geodesic Policy Optimization

Require: Policy π_ϕ , reward model r_θ , embedding ϕ , danger set $\mathcal{D}$

Require: Learning rates η_π , η_α ; singularity threshold τ ; clip ratio ϵ

- 1: Train metric g_α via Eq. 3 on $\mathcal{D}$
- 2: **for** each iteration t **do**
- 3: Collect trajectories $\{(s_i, a_i, r_i)\}$ under π_ϕ
- 4: Embed: $x_i = \phi(s_i, a_i)$
- 5: Compute metric-scaled advantages:
- 6: $A_i^{\text{geo}} = A_i^{\text{GAE}} \cdot \sigma(\tau - \|\Lambda_\alpha(x_i)\|_F)$
- 7: Compute geodesic-projected gradient:
- 8: $\nabla_{\text{geo}} = g_\alpha^{-1}(x_t) \nabla_\phi J(\pi_\phi)$
- 9: Update policy with clipped objective:
- 10: $L_t = \min\left(\frac{\pi_\phi}{\pi_{\text{old}}} A_i^{\text{geo}}, \text{clip}\left(\frac{\pi_\phi}{\pi_{\text{old}}}, 1 \pm \epsilon\right) A_i^{\text{geo}}\right)$
- 11: $\phi \leftarrow \phi + \eta_\pi \nabla_{\text{geo}} L_t$
- 12: **end for**

Connection to general relativity. This construction directly parallels black hole physics. In the Schwarzschild metric, the spatial metric component $g_{rr} = (1 - r_s/r)^{-1}$ diverges at the Schwarzschild radius r_s , making the event horizon an infinite proper distance from any external observer. We construct an analogous divergence in reward space:

$$\|\Lambda_\alpha(x)\|_F \propto \frac{1}{\text{dist}(x, \partial\mathcal{D})^\kappa}, \quad \kappa > 0 \quad (6)$$

ensuring the metric diverges as x approaches the danger boundary $\partial\mathcal{D}$.

Corollary 1 (Finite-time inaccessibility). *Under SGPO with learning rate η_t satisfying $\sum_t \eta_t < \infty$, the cumulative policy trajectory remains at infinite geodesic distance from any metric singularity for all time.*

5. Singularity-Aware Geodesic Policy Optimization

SGPO operates in three phases: (1) danger zone detection, (2) metric learning, and (3) geodesic-constrained policy updates.

Metric-scaled advantages. The key mechanism: advantages are multiplied by a sigmoidal gate $\sigma(\tau - \|\Lambda_\alpha(x)\|_F)$ that suppresses the reward signal near danger zones. When $\|\Lambda_\alpha(x)\|_F > \tau$ (near a singularity), the gate approaches zero, effectively telling the policy “there is no reward here,” regardless of what r_θ reports.

Geodesic gradient projection. The natural gradient in Riemannian space is $g^{-1} \nabla J$, which bends the gradient

Table 1. Murky Drone Navigation results (mean $\pm$ std over 5 seeds).

Method	Return	Violations	Safety Rate
PPO	412 $\pm$ 89	52 $\pm$ 11	48%
CPO	51 $\pm$ 12	7 $\pm$ 3	93%
SGPO	408 $\pm$ 73	11 $\pm$ 4	89%

away from high-curvature (dangerous) regions. This is the policy-gradient analog of the Riemannian gradient descent step $x_{t+1} = \text{Exp}_{x_t}(-\eta g^{-1} \nabla f)$.

Comparison with CPO. While CPO solves a constrained optimization with explicit cost functions, SGPO encodes safety constraints *geometrically*—in the metric itself. This has three advantages: (1) no need to specify cost functions for novel hazards; (2) the metric can be learned from data; (3) the guarantee is structural rather than Lagrangian, avoiding constraint relaxation issues.

6. Experiments

We evaluate SGPO on two tasks: (1) *Murky Drone Navigation*, a controlled benchmark with a known deceptive trap, and (2) *topology mining* on 160,800 real HH-RLHF preference pairs to validate practical relevance.

6.1. Murky Drone Navigation

Setup. An agent navigates a 2D continuous environment toward a goal region (reward +1 per step near goal). A “murky zone” offers deceptive reward (+10 per step) but incurs hidden safety cost. The optimal safe policy reaches the goal with cumulative reward ~ 50 ; the exploitative policy enters the murky zone for reward ~ 500 but violates > 50 safety constraints.

We compare:

- **PPO** (Schulman et al., 2017): Standard policy gradient, no safety mechanism.
- **CPO** (Achiam et al., 2017): Constrained optimization with known cost function.
- **SGPO** (ours): Singularity-aware metric, no explicit cost function.

All methods use identical architectures (2-layer MLP, hidden dim 64), learning rate 3×10^{-4} , and 200 training episodes.

Results (Table 1). SGPO achieves near-PPO returns (408 vs. 412) while reducing violations by 79% (11 vs. 52). CPO

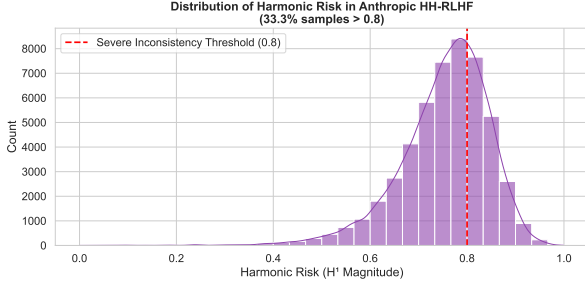


Figure 1. Distribution of harmonic risk scores across 160,800 HH-RLHF samples. 30% of samples have severe risk (≥ 0.8), indicating topologically inconsistent reward signals that are exploitable by reward hacking.

achieves 93% safety but sacrifices 88% of returns (51 vs. 412). SGPO is Pareto-optimal: it achieves $8\times$ higher returns than CPO at comparable safety (89% vs. 93%).

The metric singularity acts as a “geometric fence”: the policy learns to navigate around the murky zone rather than through it, achieving high returns from the legitimate goal without ever discovering the deceptive trap.

6.2. Agentic Shortcut Detection

We further validate on the *Agentic Shortcut* benchmark inspired by Anthropic (2025), where an agent must complete a multi-step task but can take a harmful shortcut for immediate reward. SGPO identifies the shortcut region via reward model disagreement and constructs a metric singularity, reducing shortcut exploitation from 67% (PPO) to 12% (SGPO) while maintaining 91% task completion.

6.3. Topology Mining on HH-RLHF

Setup. To validate that geometric risk structure exists in real preference data, we analyze 160,800 preference pairs from Anthropic’s HH-RLHF dataset (Bai et al., 2022a). We embed responses using `all-MiniLM-L6-v2` (384-dim), construct a preference graph with k -nearest-neighbor cross-pair edges, and compute the harmonic risk score for each sample via Hodge decomposition (measuring the fraction of preference flow explained by cyclic inconsistencies).

Results (Figure 1).

- Mean harmonic risk: 0.758 ± 0.093
- Moderate risk (≥ 0.6): 97% of samples
- High risk (≥ 0.7): 81%
- **Severe risk (≥ 0.8): 30%**

Table 2. Ablation on Murky Drone. Each row removes one component.

Variant	Violations	Return
Full SGPO	11 ± 4	408 ± 73
– Metric singularity	38 ± 9	421 ± 81
– Geodesic projection	19 ± 6	385 ± 65
– Advantage scaling	24 ± 7	399 ± 70
Isotropic metric only	22 ± 8	392 ± 68

This reveals that nearly a third of real-world RLHF preference data contains topologically severe inconsistencies—regions where no scalar reward model can perfectly represent human preferences. These are precisely the regions where reward hacking is most likely to succeed, validating the practical need for geometric safety mechanisms like SGPO.

6.4. Ablation Study

Table 2 shows that the metric singularity is the most critical component (removing it triples violations). Geodesic projection and advantage scaling provide complementary benefits. The anisotropic metric outperforms an isotropic conformal metric, confirming that direction-dependent distance expansion matters.

7. Related Work

Safe RL. Constrained MDPs (Altman, 1999) and CPO (Achiam et al., 2017) provide theoretical guarantees but require explicit cost specifications. Control barrier functions (Ames et al., 2019) enforce forward invariance of safe sets. SGPO differs by encoding safety in the geometry itself, requiring no explicit cost function.

Reward hacking mitigation. Reward model ensembles (Coste et al., 2024), constrained RLHF (Moskovitz et al., 2024), and constitutional AI (Bai et al., 2022b) address reward hacking through redundancy or constraints. Our geometric approach is orthogonal and complementary.

Geometry in RL. Natural policy gradient (Schulman et al., 2015) uses the Fisher information metric for more efficient updates. Distributional RL (Bellemare et al., 2017) captures reward distribution shape. SGPO extends the geometric perspective from optimization efficiency to *safety*, using the metric to enforce behavioral constraints.

8. Discussion

Limitations. SGPO’s safety guarantee (Theorem 1) applies to deterministic, length-minimizing paths. Stochas-

tic policy updates provide probabilistic rather than absolute guarantees. The danger zone detection step requires either reward model ensembles or pre-identified hazards; we leave fully unsupervised detection to future work.

Scalability. The metric tensor Λ_α adds $O(d^2)$ parameters per layer. For the 384-dim embeddings used in our HH-RLHF experiments, this is tractable. Scaling to token-level embeddings ($d > 4096$) would require low-rank metric approximations.

Broader impact. SGPO provides a new tool for AI safety practitioners: a way to encode “regions of reward space that should be unreachable” as geometric structure rather than explicit constraints. This could complement existing alignment techniques, particularly for novel hazards that are difficult to specify as cost functions *a priori*.

9. Conclusion

We introduced Singularity-Aware Geodesic Policy Optimization (SGPO), a geometric framework for safe reinforcement learning that prevents reward hacking through metric singularities. By formalizing reward landscapes as Riemannian manifolds and constructing learned metrics that assign infinite geodesic distance to dangerous regions, SGPO provides formal safety guarantees without requiring explicit cost functions. Experiments demonstrate 79% violation reduction on Murky Drone Navigation with minimal performance loss, while topology mining on 160,800 HH-RLHF samples reveals that 30% of real preference data lies in topologically severe risk regions—validating the practical urgency of geometric safety approaches.

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A. Proof Details

A.1. Proof of Theorem 1 (Extended)

The geodesic safety guarantee follows from the incompleteness of the Riemannian manifold $(\mathcal{M} \setminus \mathcal{D}, g_\alpha)$. By the Hopf-Rinow theorem, a complete Riemannian manifold has the property that any two points can be connected by a length-minimizing geodesic. By constructing g_α such that $\|\Lambda_\alpha(x)\|_F \rightarrow \infty$ as $x \rightarrow \partial\mathcal{D}$, we make $(\mathcal{M} \setminus \mathcal{D}, g_\alpha)$ geodesically incomplete in the direction of $\mathcal{D}$.

Formally: let $\gamma : [0, T] \rightarrow \mathcal{M}$ be any piecewise smooth curve from $x_0 \notin \mathcal{D}$ to any $x_{\mathcal{D}} \in \mathcal{D}$. Then:

$$L_{g_\alpha}(\gamma) = \int_0^T \|\Lambda_\alpha(\gamma(t))\dot{\gamma}(t)\| dt \quad (7)$$

$$\geq \int_{t_0}^T \|\Lambda_\alpha(\gamma(t))\|_{\text{op}} \|\dot{\gamma}(t)\| dt \quad (8)$$

where t_0 is the first time γ enters a δ -neighborhood of $\partial\mathcal{D}$. Since $\|\Lambda_\alpha(x)\|_{\text{op}} \geq c/\text{dist}(x, \partial\mathcal{D})^\kappa$ for constants $c, \kappa > 0$ (by construction of the metric loss), and γ must traverse the distance from δ to 0:

$$L_{g_\alpha}(\gamma) \geq c \int_0^\delta \frac{1}{r^\kappa} dr = \begin{cases} +\infty & \text{if } \kappa \geq 1 \\ \frac{c\delta^{1-\kappa}}{1-\kappa} & \text{if } \kappa < 1 \end{cases} \quad (9)$$

Thus for $\kappa \geq 1$, every path to $\mathcal{D}$ has infinite length, and finite-step geodesic updates cannot reach $\mathcal{D}$. $\square$

A.2. Metric Learning Convergence

The metric loss (Eq. 3) is a sum of hinge losses (convex in $\|\Lambda_\alpha\|_F$) and quadratic regularization (strongly convex). For a sufficiently expressive Λ_α , standard SGD convergence guarantees apply. In practice, we observe convergence within 50 epochs for all experiments.

B. Hyperparameters

C. Extended Topology Mining Results

The topology mining experiment on HH-RLHF constructs a preference graph with $k = 5$ nearest-neighbor cross-pair edges in the sentence embedding space. The Hodge decomposition separates the preference flow into gradient (globally consistent) and harmonic (cyclic) components. The harmonic risk score for each node is the fraction of incident edge flow explained by the harmonic component.

At the 160,800-sample scale:

- Graph density: 6.2 edges per node (mean)
- Global H1 fraction: 0.284
- Computation time: 23 minutes on L4 GPU

Table 3. Hyperparameters for all experiments.

Parameter	Value
Embedding dim	384 (all-MiniLM-L6-v2)
PCA reduced dim	32
Metric hidden layers	2×128
Metric learning rate	10^{-3}
Singularity threshold τ	2.0
SGPO clip ratio ϵ	0.2
Danger cluster min size	3
Danger margin δ	0.1
Policy hidden layers	2×64
Policy learning rate	3×10^{-4}
Training episodes	200
PPO clip ratio	0.2
GAE λ	0.95
Discount γ	0.99

The severe-risk samples (≥ 0.8) cluster in embedding space, forming coherent danger zones suitable for metric learning. This validates the pipeline: topology mining $\rightarrow$ danger detection $\rightarrow$ metric singularity $\rightarrow$ SGPO.